Learn the Art of Linux Troubleshooting

Here are some valuable tips to help you find important system files in RHEL 5, which get deleted by accident.

Everything seems fine when your Linux machine works just the way you want it to. But that feeling changes dramatically when your machine starts creating problems that you find really difficult to sort out. Not everyone can troubleshoot a Linux machine efficiently… but you can, if you are ready to stay with us for the next few minutes. Let's look at what to do when some of your important system files are deleted or corrupted under Red Hat Enterprise Linux 5. The action begins now!

Scene 1: /etc/passwd is deleted

This is an important file in Linux as it contains information about user accounts and passwords. If it's missing in your system and you try to log in to a user account, you get an error message stating 'Log-in incorrect' and after restarting the system, you will see the screen shot shown in Figure 1.

Now that you have seen the problem and its consequences, it's time to solve it. Boot into single user mode. At the start of booting, press any key to enter into the GRUB menu. Here you will see a list of the operating systems installed. Just select the one you are working with and press 'e'.

It's time to have some fun with kernel parameters. So highlight the kernel and again press 'e' to edit its parameters.

Next, instruct the kernel to boot into single user mode, which is also known as maintenance mode. Just type *'1'* after a space and press the *Enter* key. Now press *'b'* to continue the booting process.

Now that you have booted into single user mode, you are probably asking yourself, "What is next?". The tricky portion of this exercise is now over and it takes just one command to have your *passwd* file in its place. Actually, there is a file */etc/passwd-*, which is nothing but the backup file for */etc/passwd*. So all you need to do is to issue the following command:

```
cp /etc/passwd- /etc/passwd
```

…and you are done. Now you can issue the *init 5* command to switch to the graphical mode. Everything is fine now. You can also find the backup of */etc/shadow* and */etc/gshadow as /etc/shadow-* and */etc/gshadow-* respectively.

Scene 2: /etc/pam.d/login is deleted

If your */etc/pam.d/login* file is deleted and you try to log in, it won't ask you to enter your password after entering your username. Instead, it will continuously show the localhost login prompt. Here again, there is a single command that will solve the problem for you:

```
cp /etc/pam.d/system-auth /etc/pam.d/login
```

Just boot into the single user mode as done earlier, type this command and you'll be able to log in normally. There is also a second solution to this problem, which we'll look at after a while.